PROJECT: A SIMPLE MACHINE LEARNING PIPELINE WITH R SHINY

GROUP G

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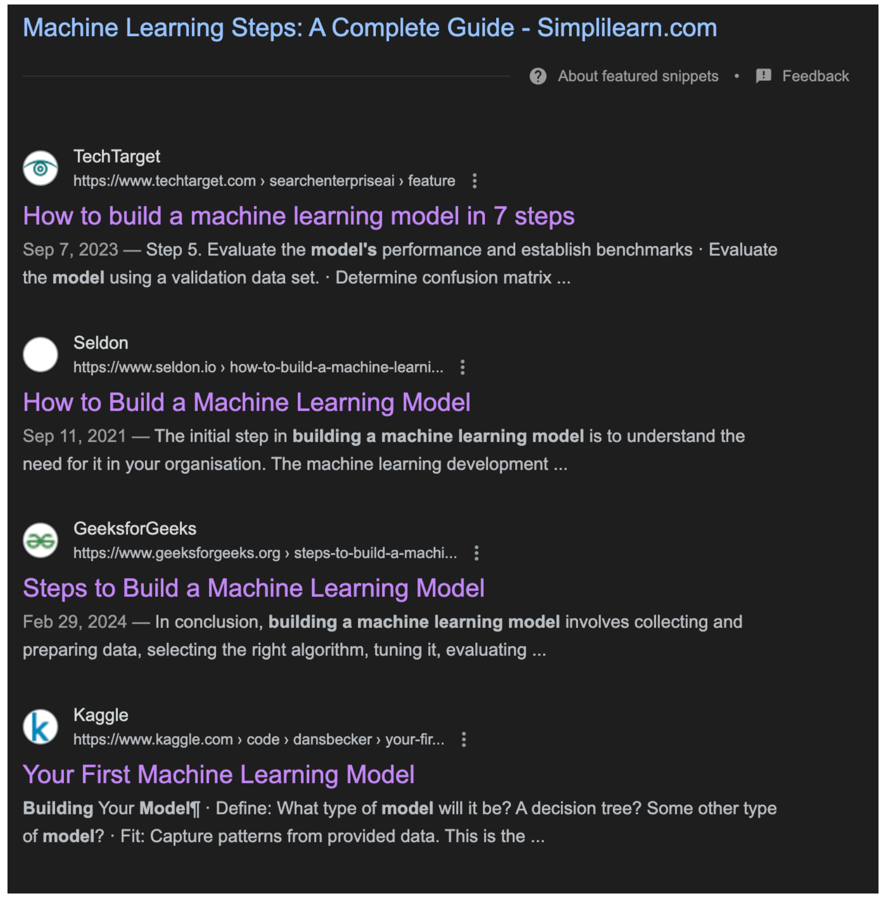
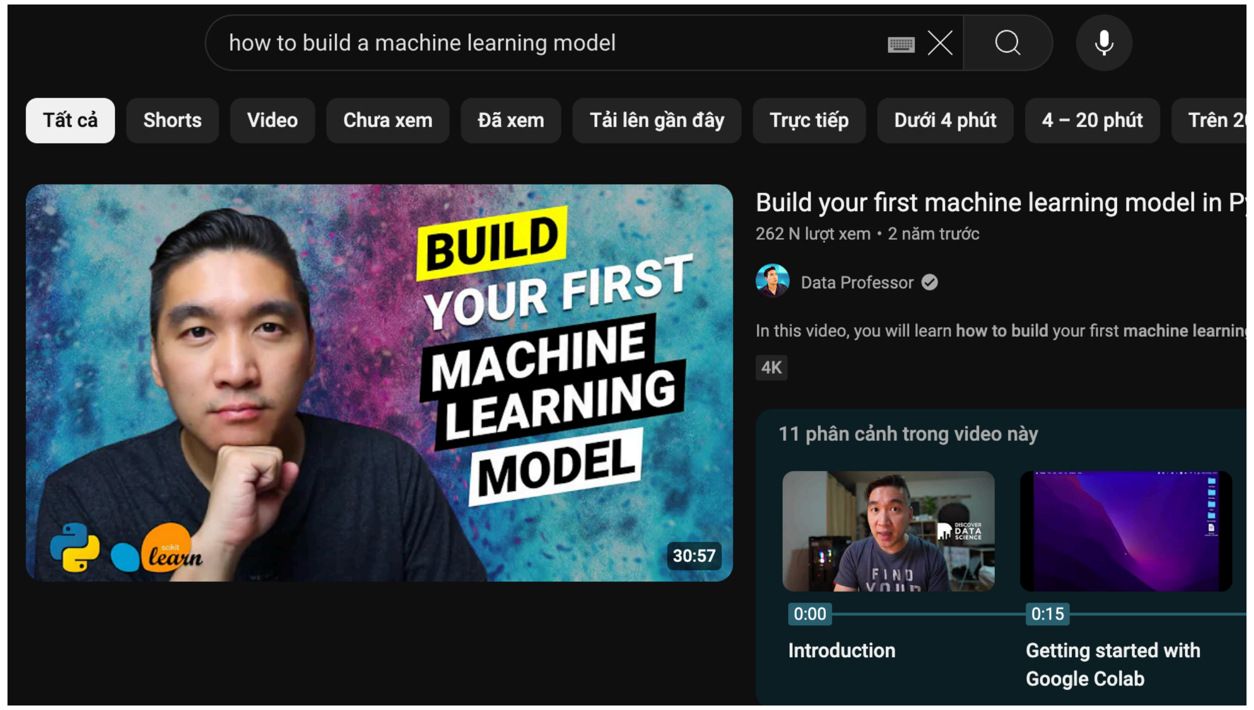
MOTIVATION

Technical knowledge and setup barriers prevent non-technical users from exploring machine learning:

- **Technical knowledge**: Traditional machine learning platforms require programming knowledge and familiarity with complex algorithms.
- **Technical set-up**: Setting up the necessary environments and dependencies can be time-consuming and challenging for beginners, unlike analyzing data by tools such as Excel or **PowerBI.**





OUR SOLUTION

We create a platform that **simplifies the process of building and understanding machine learning models.**

Our platform leverages visualization and interactivity to provide an intuitive machine learning experience without coding.

DATASET DESCRIPTION

IRIS DATASET

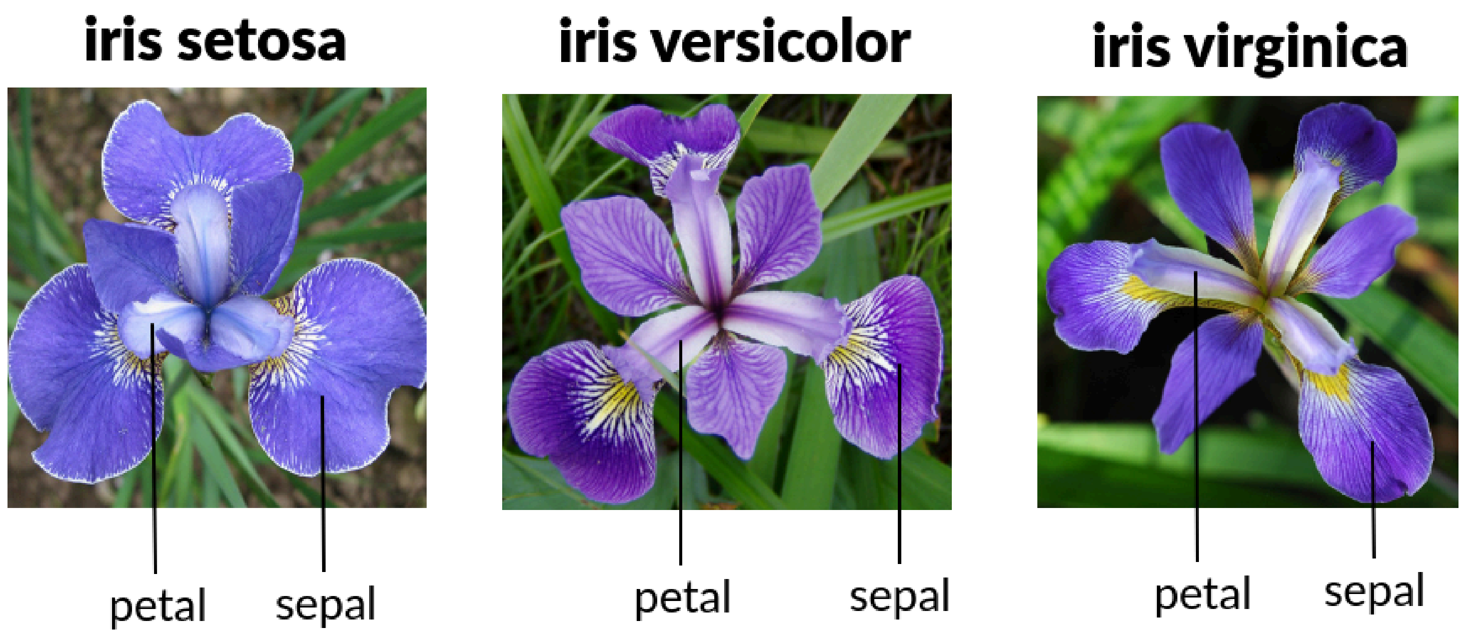
Iris dataset: **150 samples of iris flowers.**

Four features were measured:

- **Sepal Length**: Length of the sepals (the green leaf-like structures that protect the flower bud) in centimeters.
- **Sepal Width**: Width of the sepals in centimeters.
- **Petal Length**: Length of the petals (the colored part of the flower) in centimeters.
- **Petal Width**: Width of the petals in centimeters.

Target variable: species of each iris flower:

Setosa, Versicolor, Virginica



BY PRAMOD SAHU

OUR PRODUCT

A machine learning pipeline with different tabs, each representing a step in the pipeline

The tabs in the products are:

- Data Upload: User will upload a chosen dataset
- Data Summary: Give an overview by providing a data summary and a simple distribution plot
- Data Exploration: Preprocess data by deleting/filling NAs and creating simple plots
- Build Model: Pick variables, choose algorithms

We also incorporate a tab on reducing high dimensional data by using T-SNE

DATA UPLOAD

Upload Your Dataset

Upload data (CSV format recommended)

Browse... sample.csv

Upload complete

Data Uploaded:

Name	Size
sample.csv	818

Data Preview:

Show 10 entries

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
All	All	All	All	All
5.1	3.5	1.4	0.2	Setosa
4.9	3	1.4	0.2	Setosa
4.7	3.2	1.3	0.2	Setosa
4.6	3.1	1.5	0.2	Setosa
5	3.6	1.4	0.2	Setosa
5.4	3.9	1.7	0.4	Setosa
4.6	3.4	1.4	0.3	Setosa
5	3.4	1.5	0.2	Setosa
4.4	2.9	1.4	0.2	Setosa
4.9	3.1	1.5	0.1	Setosa

Showing 1 to 10 of 30 entries

Previous 1 2 3 Next

DATA SUMMARY

Data Summary

This section provides a summary of the uploaded data.

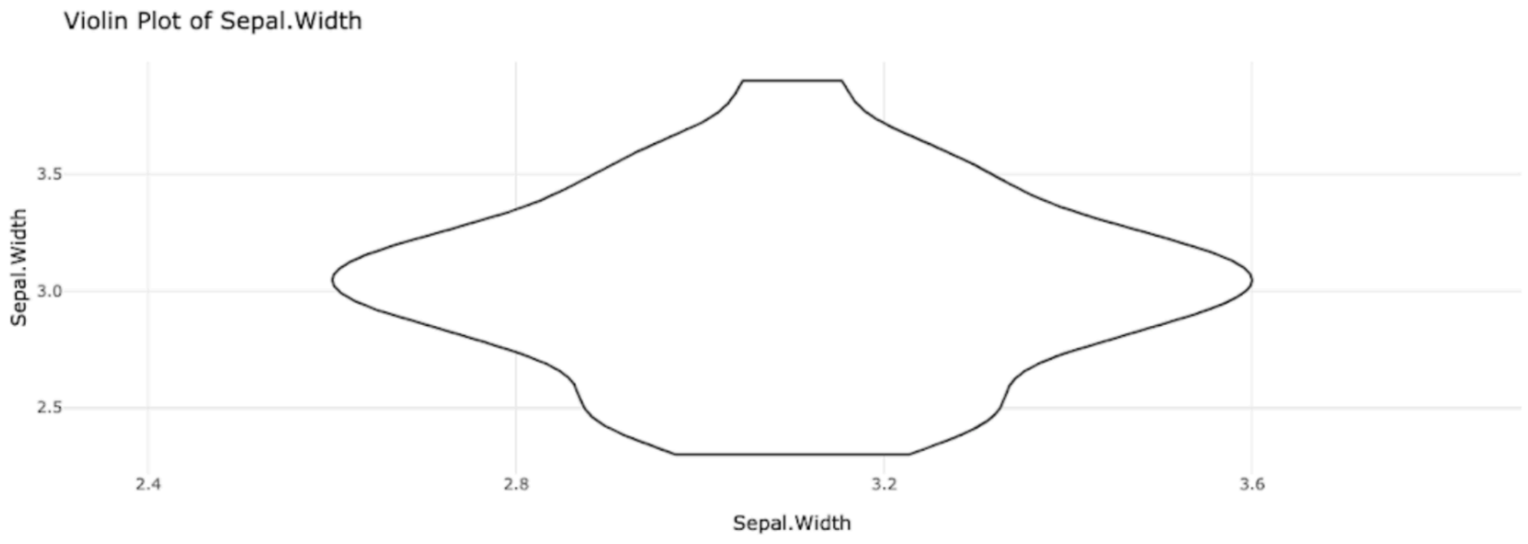
Select variable for violin plot:

Sepal.Width

Summary of Data

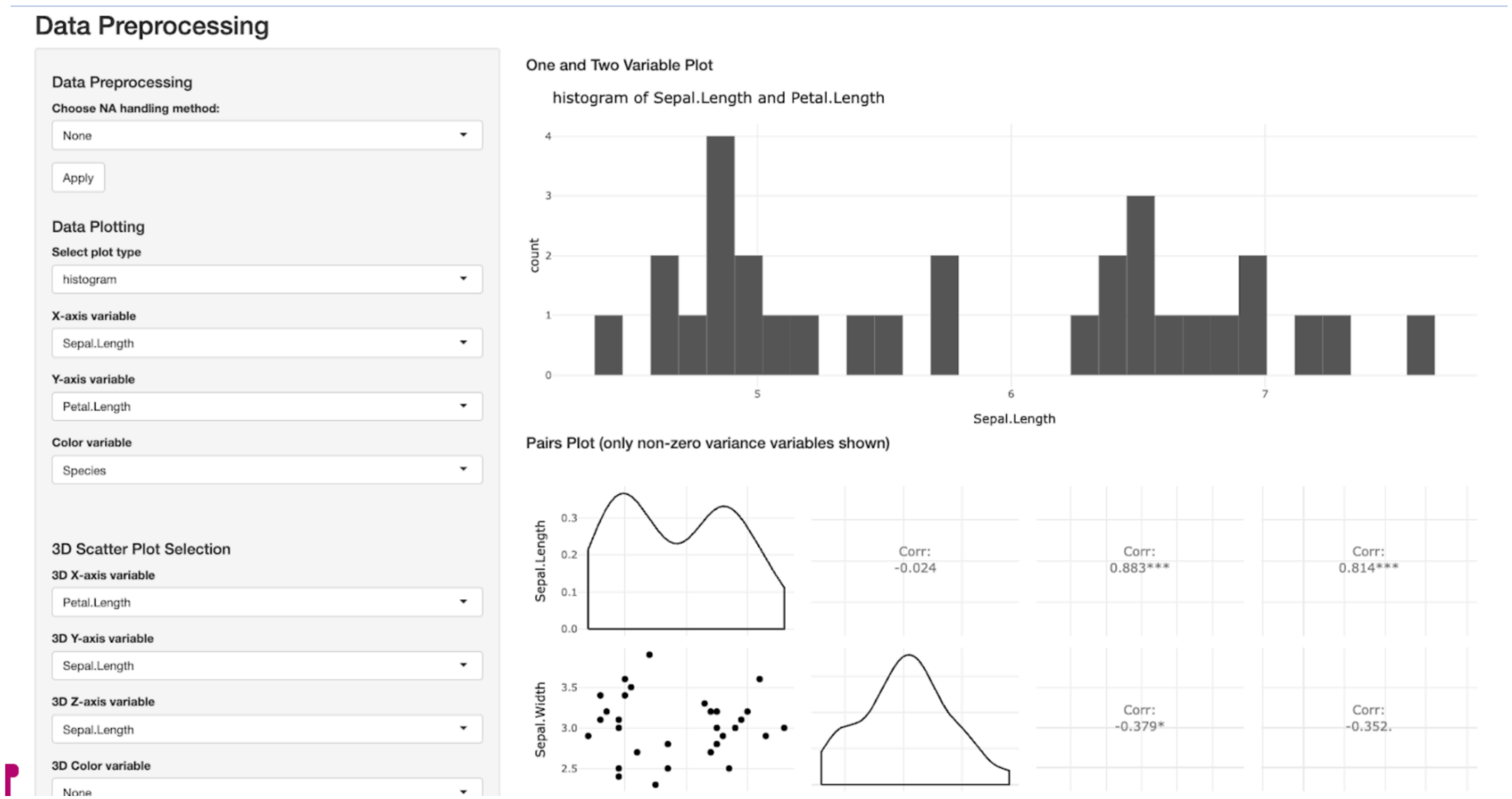
	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
Sepal.Length	1	30.00	5.84	0.96	5.70	5.81	1.19	4.40	7.60	3.20	0.13	-1.48	0.17
Petal.Length	2	30.00	3.81	1.83	4.50	3.80	1.85	1.30	6.60	5.30	-0.27	-1.53	0.33
Petal.Width	3	30.00	1.20	0.77	1.40	1.20	0.89	0.10	2.50	2.40	-0.22	-1.48	0.14
Species*	4	30.00	2.00	0.83	2.00	2.00	1.48	1.00	3.00	2.00	0.00	-1.60	0.15

Violin Plot



- Allow user to observe a summary of their dataset: mean, median, standard deviations, etc.
- Violin plot to help user look at the distribution of each variable

DATA EXPLORATION



- Allow user to observe a summary of their dataset: mean, median, standard deviations, etc.
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BUILD MODEL

LIMITATIONS

- **Educational depth vs. usability trade-off**:
    - The platform's primary goal is to be user-friendly and accessible >< reduced educational depth.
    - Although we have included some tutorials for in depth explanation, we could only compromise for a high level understanding.
- **Restricted flexibility:**
    - Does not provide the flexibility required for users to deeply engage with and understand machine learning concepts and techniques.
    - Falls short in allowing hands-on practice

FUTURE DIRECTION

- **Enhanced customization and interactivity**:
  - Expand the parameter and configuration options available
  - Integrate a code editor or Notebook-style interface to enable direct interaction
  - Enhance the data visualization capabilities with interactive plots and charts

- **Support broader range of data format:** Currently we only adopt a dataset based on the Iris dataset -> more formats should be supported for the sake of users' experience.

- **Incorporate model evaluation**: One of the major future update for our app, as this is an important step in the pipeline

- **Incorporate more classic algorithms**: In real life, there are more classic, but important models that we barely touch on, for example, linear regression.

CONCLUSION

- We developed a user-friendly machine learning pipeline using R Shiny
- Future iterations aim to address these limitations by: incorporating enhanced customization, a wider range of data format support, and improved multi-dimensional data visualization.
- We hope these advancements will further democratize machine learning education

THANK YOU